The Cannonball Circus Caper

Whodunnit? · Quadratics Name: _____

Hush fell over the Big Top as the Ringmaster reached for the **Golden Top Hat**, the circus's most famous prize, to start the grand finale. But the hat stand was empty!

Six performers were at the circus, each in a different place, and there are six ways to sneak a hat out of the tent. You are the lead detective: find **who** took it, **where** they were, and **how** they did it.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Ringmaster Lopez Circus Boss	☐ Big Top Ring	☐ Fired from the cannon
☐ Zara Acrobat	☐ Trapeze Platform	☐ Squeezed in a clown car
☐ Bumbles Clown	☐ Clown Car Garage	☐ Hidden in a popcorn tub
☐ Strongman Gus Weightlifter	☐ Popcorn Stand	☐ Rolled out in a barrel
☐ Madame Fortuna Fortune Teller	☐ Animal Tent	☐ Stuffed in a drum
☐ Trainer Ines Animal Trainer	☐ Cannon Platform	☐ Carried in a juggler's bag

The culprit is _____ in _____ using _____

Show Your Work

The Cannonball Circus Caper · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Cannonball Circus Caper

Solve by factoring:

$$x^2 - 5x + 6 = 0$$

A. $x = -2, -3$	Bumbles was honking his horn in the Big Top Ring. The front row can't stop laughing about it.
B. $x = 1, 6$	Zara was checking her helmet on the Cannon Platform before her act. A stagehand chatted with her the whole time.
C. $x = 5, 6$	The cannon's barrel was checked and is spotless. Nothing has been fired from it today.
D. $x = 2, 3$	Ringmaster Lopez was announcing the next act in the Big Top Ring. The whole audience heard him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Cannonball Circus Caper

Solve by factoring:

$$x^2 + 5x + 6 = 0$$

A. $x = -1, -6$	Strongman Gus was hauling a heavy sandbag onto the Cannon Platform. Two stagehands counted his trips.
B. $x = -2, -3$	The clown car was searched. Only 14 rubber chickens inside.
C. $x = 2, 3$	Bumbles was juggling popcorn at the Popcorn Stand. The vendor has the buttery fingerprints to prove it.
D. $x = -5, -6$	The cannon's fuse is still wrapped in plastic. It was never lit.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Cannonball Circus Caper

Solve by factoring:

$$x^2 + x - 6 = 0$$

A. $x = -1, 6$	The cannon's safety pin is locked in place. It could not have fired today.
B. $x = 2, -3$	Zara was swinging on the Trapeze Platform with her safety line recorded on camera.
C. $x = -2, 3$	Bumbles was making balloon animals in the Animal Tent. The pony wore one as a hat.
D. $x = 1, -6$	Madame Fortuna was reading palms on the Cannon Platform. The cannon crew got very good fortunes.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #4

The Cannonball Circus Caper

Solve by factoring:

$$x^2 - x - 6 = 0$$

A. $x = 2, -3$	Bumbles was holding the net rope on the Trapeze Platform. The safety crew signed off on it.
B. $x = 1, -6$	The cannon crew swept the barrel clean and found nothing but one dusty piece of confetti.
C. $x = -1, 6$	Trainer Ines was calming a nervous pony on the Cannon Platform. The crowd below clapped for her.
D. $x = -2, 3$	Every popcorn tub was emptied and counted. No hat.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Cannonball Circus Caper

Solve:

$$x^2 = 16$$

A. $x = \pm 4$	Strongman Gus was lifting a pretzel for a kid at the Popcorn Stand. The vendor remembers.
B. $x = \pm 2$	The cannon's powder tank is full to the brim. Not a drop was used.
C. $x = \pm 8$	Ringmaster Lopez was giving the safety speech on the Cannon Platform. The loudspeaker log has the time.
D. $x = 4$	Bumbles was squeezing into the tiny car in the Clown Car Garage for practice. Five clowns watched and counted.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #6

The Cannonball Circus Caper

Solve:

$$2x^2 = 50$$

A. $x = 5$	Zara was tying her sparkly boots on the Cannon Platform. The photographer snapped a picture of it.
B. $x = \pm\sqrt{50}$	The net below the cannon is dry and unmarked. Nobody landed in it today.
C. $x = \pm 5$	The barrel by the gate was rolled open. Just old costumes inside.
D. $x = \pm 25$	Bumbles was tripping over his big shoes in the Big Top Ring. The audience gasped and laughed.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Cannonball Circus Caper

Use the quadratic formula to solve:

$$x^2 - 6x + 4 = 0$$

A. $x = -3 \pm \sqrt{5}$	Bumbles was squirting seltzer at the Popcorn Stand. The vendor still has a wet apron.
B. $x = 3 \pm \sqrt{10}$	The cannon's launch counter reads zero. It hasn't gone off since last week's show.
C. $x = 3 \pm 2\sqrt{5}$	Strongman Gus was tightening bolts on the Cannon Platform. The crew chief signed his work order.
D. $x = 3 \pm \sqrt{5}$	Madame Fortuna was telling fortunes in the Animal Tent. Her appointment book shows it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #8

The Cannonball Circus Caper

Find the vertex of $y = x^2 - 4x + 1$

A. (4, 1)	The cannon crew swears it stayed covered all evening. The cover's zipper is still sealed.
B. (2, −3)	The drum was opened. It only holds drumsticks.
C. (−2, 13)	Bumbles was waving from the Trapeze Platform in his rainbow wig. The crowd below spotted him right away.
D. (2, 1)	Madame Fortuna was waving her scarf at the Cannon Platform to bless the show. A dozen people watched her.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Cannonball Circus Caper

Solve by factoring:

$$x^2 - 2x - 8 = 0$$

A. $x = 4, -2$	Trainer Ines was washing the clown car in the Clown Car Garage with two helpers.
B. $x = 8, -1$	Ringmaster Lopez was testing the spotlight from the Cannon Platform. The lighting crew confirmed it.
C. $x = -4, 2$	Bumbles was teaching a parrot to honk in the Animal Tent. The trainer's notebook mentions it.
D. $x = 4, 2$	Soot marks inside the cannon are old and faded. Nothing has been fired lately.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #10

The Cannonball Circus Caper

Find the equation of the axis of symmetry of $y = x^2 + 6x + 5$

A. $x = -6$	Bumbles was repainting the clown car's polka dots in the Clown Car Garage. His paint-covered shoes are still there.
B. $x = -5$	A stagehand checked the cannon with a flashlight. It was empty from end to end.
C. $x = -3$	The juggler’s bag was checked. Only bowling pins and one banana.
D. $x = 3$	Trainer Ines was posing with a tiger cub on the Cannon Platform. A kid's photo shows the time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.